

Thm. Riemann's Theorem.

Let f be a complex function having an isolated singularity at $z = z_0$.

If f is bounded for some deleted neighborhood of z_0 ,

then z_0 is removable.

Proof. By the hypothesis, for some $R > 0$, f is analytic and bounded in

$$0 < |z - z_0| < R.$$

Given $z_1 \in B_R(z_0) \setminus \{z_0\}$, let $r > 0$ s.t. $r < |z_1 - z_0| < R$. So, f is analytic and bounded in the annulus between $C_1: |z - z_0| = R$ and $C_2: |z - z_0| = r$.

By the Cauchy's integral formula,

$$f(z_1) = \frac{1}{2\pi i} \cdot \left[\int_{C_1} \frac{f(u)}{u - z_1} du - \int_{C_2} \frac{f(u)}{u - z_1} du \right]$$

Meanwhile, by the ML inequality,

$$\left| \int_{C_2} \frac{f(u)}{u - z_1} du \right| \leq \sup |f| \cdot 2\pi r \cdot \frac{1}{|z_1 - z_0| - r} \rightarrow 0 \text{ as } r \rightarrow 0.$$

$$\text{So, } f(z_1) = \frac{1}{2\pi i} \int_{C_1} \frac{f(u)}{u - z_1} du.$$

Since z_1 was arbitrary, we can denote that

$$f(z) = \frac{1}{2\pi i} \cdot \int_{C_1} \frac{f(u)}{u - z} du \quad \text{for all } 0 < |z - z_0| < R,$$

$$\text{So, define } f(z_0) = \frac{1}{2\pi i} \cdot \int_{C_1} \frac{f(u)}{u - z_0} du, \quad \text{we obtain the result.}$$

[Thm. Suppose that f has an isolated singularity at z_0 . TFAE:

1) For some $m \in \mathbb{N}$, $\lim_{z \rightarrow z_0} (z - z_0)^m \cdot f(z) = A \in \mathbb{C} \setminus \{0\}$. (That is, z_0 is pole of order m).

2). $f(z) \rightarrow \infty$ as $z \rightarrow z_0$

3). $f(z) = g(z)/(z - z_0)^m$ for some analytic $g(z)$ and $m \in \mathbb{N}$, with $g(z_0) \neq 0$.

Proof. Suppose that there exists $m \in \mathbb{N}$ s.t. $\lim_{z \rightarrow z_0} (z - z_0)^m f(z)$ exists with non-zero.

Given $\varepsilon = |A|/2$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \Rightarrow |(z - z_0)^m \cdot f(z) - A| < |A|$$

So, using the triangle inequality, for $0 < |z - z_0| < \delta$,

$$|A| - |z - z_0|^m |f(z)| < \frac{|A|}{2}$$

$$\Rightarrow \frac{|A|}{2\delta^m} < \frac{|A|}{2|z - z_0|^m} < |f(z)|$$

Thus, taking $\delta \rightarrow 0$, then $f(z) \rightarrow \infty$. [1) $\Rightarrow$ 2).

Suppose that $f(z) \rightarrow \infty$ as $z \rightarrow z_0$. Put $g(z) = 1/f(z)$ in $B_\delta(z_0) \setminus \{z_0\}$ for some $\delta > 0$, which satisfies $f(z) \neq 0$. (If no such neighbor exists, then f is identically zero.)

Now, since $f(z) \rightarrow \infty$ as $z \rightarrow z_0$, so $g(z) \rightarrow 0$ as $z \rightarrow z_0$, so z_0 is removable to g .

Then, the Taylor expansion

$$g(z) = a_0 + a_1(z - z_0) + \dots \quad (|z - z_0| < \delta).$$

is valid. Since $g(z) \neq 0$ in $B_\delta(z_0) \setminus \{z_0\}$, so not all the coefficients are zero.

Let m be the smallest integer s.t. $a_m \neq 0$. ($m \geq 1$ because $\lim_{z \rightarrow z_0} g(z) = a_0 = 0$).

So that $g(z) = a_m(z - z_0)^m + a_{m+1}(z - z_0)^{m+1} + \dots$

Now, $\frac{1}{(z - z_0)^m f(z)} = a_m + a_{m+1}(z - z_0) + \dots \rightarrow a_m$ as $z \rightarrow z_0$,

that is, $(z - z_0)^m f(z) \rightarrow a_m^{-1}$ as $z \rightarrow z_0$. [2) $\Rightarrow$ 3).

3) $\Rightarrow$ 2) is directly, the analytic g is continuous at z_0 .

Isolated singularity at ∞ .

If f is analytic for all $R < |z| < \infty$, for some $R > 0$, then we say that f has an isolated singularity at ∞ .

For this $R > 0$, observe that

$f(z)$ is analytic at $|z| > R$ if and only if $f(\frac{1}{z})$ is analytic at $|z| < \frac{1}{R}$.

$f(z)$ has an isolated singularity at ∞ iff $f(\frac{1}{z})$ has an isolated singularity at 0.

Thm. Let f be a non-constant entire function. Then,

f is polynomial if and only if $f(z) \rightarrow \infty$ as $z \rightarrow \infty$.

Proof. Suppose that f is a polynomial. WLOG, let $f(z) = a_0 + a_1 z + \dots + a_n z^n$. ($a_n \neq 0$).

Then, $f(\frac{1}{z}) = \frac{a_n}{z^n} + \frac{a_{n-1}}{z^{n-1}} + \dots + \frac{a_1}{z} + a_0$ so that $\lim_{z \rightarrow 0} z^n \cdot f(\frac{1}{z}) = a_n \neq 0$.

So, $f(z)$ has a pole at ∞ , so $\lim_{z \rightarrow \infty} f(z) = \infty$.

Conversely, suppose that: $f(\frac{1}{z})$ has a pole of order m at 0. That is,

$$\lim_{z \rightarrow 0} \frac{f(z)}{z^m} = A \in \mathbb{C} \setminus \{0\}.$$

That is, for some $R > 0$, $|z| \geq R \Rightarrow \left| \frac{f(z)}{z^m} - A \right| < 1$

$$\Rightarrow \left| \frac{f(z)}{z^m} \right| < 1 + |A|$$

$$\Rightarrow |f(z)| < (1 + |A|) \cdot z^m$$

So, f is a polynomial of degree at most m .

Singularities of rational functions.

Let P and Q be polynomials over $\mathbb{C}$, $P(z) = a_0 + a_1 z + \dots + a_n z^n$, $Q(z) = b_0 + \dots + b_m z^m$
and $f(z) = P(z)/Q(z)$. Then, $(a_n, b_m \neq 0)$.

- $f(z)$ has a removable singularity at ∞
- iff $f(1/z)$ has a removable singularity at 0
- iff for some $\epsilon > 0$ and $M > 0$, $0 < |1/z| < \epsilon \Rightarrow |f(1/z)| \leq M$.
- iff $|z| > \frac{1}{\epsilon} \Rightarrow |f(z)| \leq M$.
- iff $\deg P(z) \leq \deg Q(z)$.

- $f(z)$ has a pole of order k at ∞
- iff $f(1/z)$ has a pole of order k at 0.
- iff $f(1/z) = \frac{P(1/z)}{Q(1/z)} = \sum_{i=-k}^{\infty} C_i \cdot z^i$ for some $\{C_i\}_{i=-k}^{\infty}$

iff $P(1/z) = \sum_{i=0}^n a_i \cdot \frac{1}{z^i} = \left(\sum_{i=0}^m b_i \cdot \frac{1}{z^i} \right) \cdot \left(\sum_{i=-k}^{\infty} C_i z^i \right)$

- iff $n = m + k$, that is, the order $k = \deg P - \deg Q > 0$.

Long division.

Consider a function represented by

$$f(z) = \left(\sum_{n=0}^{\infty} a_n z^n \right) / \left(\sum_{n=0}^{\infty} b_n z^n \right) = \frac{a_0 + a_1 z + a_2 z^2 + \dots}{b_0 + b_1 z + b_2 z^2 + \dots} \quad (b_0 \neq 0)$$

Since $b_0 \neq 0$, this f is analytic at $z=0$, so f can be written as

$$f(z) = \frac{a_0 + a_1 z + \dots}{b_0 + b_1 z + \dots} = c_0 + c_1 z + \dots$$

so that

$$\left(\sum_{n=0}^{\infty} a_n z^n \right) = \left(\sum_{n=0}^{\infty} b_n z^n \right) \cdot \left(\sum_{n=0}^{\infty} c_n z^n \right), \text{ in some neighbor of the origin, thus}$$

the uniqueness theorem gives

$$a_n = \sum_{k=0}^n b_k \cdot c_{n-k}, \text{ the Cauchy product.}$$

Example.

Let $f(z) = \frac{\pi \cdot \cot \pi z}{z^4}$. Then,

$$\begin{aligned} f(z) &= \frac{\pi}{z^4} \cdot \left(\frac{1 - \frac{\pi^2}{2!} z^2 + \frac{\pi^4}{4!} z^4 + \dots}{\pi z - \frac{\pi^3}{3!} z^3 - \frac{\pi^5}{5!} z^5 + \dots} \right) \\ &= \frac{1}{z^5} \cdot \left(\frac{1 - \frac{\pi^2}{2} z^2 + \frac{\pi^4}{4!} z^4 - \dots}{1 - \frac{\pi^2}{3!} z^2 + \frac{\pi^4}{5!} z^4 - \dots} \right) \end{aligned}$$

$$1 = 1 \cdot c_0 \Rightarrow c_0 = 1$$

$$a_1 = b_0 c_1 + b_1 c_0 = c_1 + 0 = 0. \Rightarrow c_1 = 0.$$

$$a_2 = b_0 c_2 + b_1 c_1 + b_2 c_0 = c_2 + 0 + -\frac{\pi^2}{8} \cdot 1 = -\frac{\pi^2}{8} \Rightarrow c_2 = -\frac{\pi^2}{8}.$$

$$a_3 = b_0 c_3 + b_1 c_2 + b_2 c_1 + b_3 c_0 = c_3 + 0 + -\frac{\pi^2}{8} \cdot 0 + 0 = 0 \Rightarrow c_3 = 0.$$

$$a_4 = b_0 c_4 + b_1 c_3 + b_2 c_2 + b_3 c_1 + b_4 c_0 = c_4 + 0 - \frac{\pi^2}{8} + \frac{\pi^4}{5!} = c_4 + \pi^4 \cdot \left(\frac{1}{18} + \frac{1}{120} \right) = \frac{\pi^4}{24}.$$

$$\Rightarrow c_4 = \pi^4 \left(\frac{1}{24} - \frac{1}{18} - \frac{1}{120} \right) = \pi^4 \cdot \left(-\frac{1}{45} \right) = -\frac{1}{45} \pi^4.$$

Consequently, the principal part of f is

$$\frac{1}{z^5} - \frac{\pi^2}{3} \cdot \frac{1}{z^3} - \frac{\pi^4}{45} \cdot \frac{1}{z}.$$